

Differentiation-Continuous Functions

Numerical methods

Forward Difference Approximation

$$f'(x) = \lim_{\Delta x \rightarrow 0} \frac{f(x + \Delta x) - f(x)}{\Delta x}$$

For a finite ' Δx '

$$f'(x) \approx \frac{f(x + \Delta x) - f(x)}{\Delta x}$$

Graphical Representation Of Forward Difference Approximation

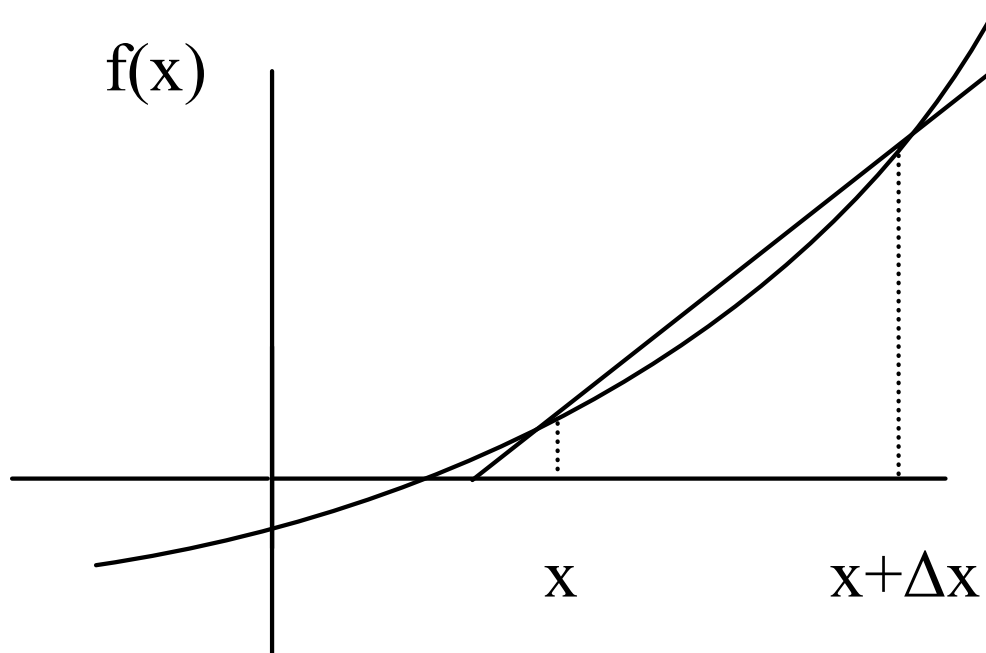


Figure 1 Graphical Representation of forward difference approximation of first derivative.

Example 1

The velocity of a rocket is given by

$$v(t) = 2000 \ln \left[\frac{14 \times 10^4}{14 \times 10^4 - 2100t} \right] - 9.8t, 0 \leq t \leq 30$$

where ' v ' is given in m/s and ' t ' is given in seconds.

- Use forward difference approximation of the first derivative of $v(t)$ to calculate the acceleration at $t = 16s$. Use a step size of $\Delta t = 2s$.
- Find the exact value of the acceleration of the rocket.
- Calculate the absolute relative true error for part (b).

Example 1 Cont.

Solution

$$a(t_i) \approx \frac{v(t_{i+1}) - v(t_i)}{\Delta t}$$

$$t_i = 16$$

$$\Delta t = 2$$

$$\begin{aligned} t_{i+1} &= t_i + \Delta t \\ &= 16 + 2 \\ &= 18 \end{aligned}$$

$$a(16) \approx \frac{v(18) - v(16)}{2}$$

Example 1 Cont.

$$\begin{aligned}v(18) &= 2000 \ln \left[\frac{14 \times 10^4}{14 \times 10^4 - 2100(18)} \right] - 9.8(18) \\&= 453.02 \text{ m/s}\end{aligned}$$

$$\begin{aligned}v(16) &= 2000 \ln \left[\frac{14 \times 10^4}{14 \times 10^4 - 2100(16)} \right] - 9.8(16) \\&= 392.07 \text{ m/s}\end{aligned}$$

Hence

$$a(16) \approx \frac{v(18) - v(16)}{2}$$

Example 1 Cont.

$$\approx \frac{453.02 - 392.07}{2}$$
$$\approx 30.474 \text{ m/s}^2$$

b) The exact value of $a(16)$ can be calculated by differentiating

$$v(t) = 2000 \ln \left[\frac{14 \times 10^4}{14 \times 10^4 - 2100t} \right] - 9.8t$$

as

$$a(t) = \frac{d}{dt} [v(t)]$$

Example 1 Cont.

Knowing that

$$\frac{d}{dt}[\ln(t)] = \frac{1}{t} \quad \text{and} \quad \frac{d}{dt}\left[\frac{1}{t}\right] = -\frac{1}{t^2} \quad v(t) = 2000 \ln\left[\frac{14 \times 10^4}{14 \times 10^4 - 2100t}\right] - 9.8t$$

$$a(t) = 2000 \left(\frac{14 \times 10^4 - 2100t}{14 \times 10^4} \right) \frac{d}{dt} \left(\frac{14 \times 10^4}{14 \times 10^4 - 2100t} \right) - 9.8$$

$$= 2000 \left(\frac{14 \times 10^4 - 2100t}{14 \times 10^4} \right) (-1) \left(\frac{14 \times 10^4}{(14 \times 10^4 - 2100t)^2} \right) (-2100) - 9.8$$

$$= \frac{-4040 - 29.4t}{-200 + 3t}$$

Example 1 Cont.

$$a(16) = \frac{-4040 - 29.4(16)}{-200 + 3(16)}$$
$$= 29.674 \text{ m/s}^2$$

The absolute relative true error is

$$|\epsilon_t| = \left| \frac{\text{True Value} - \text{Approximate Value}}{\text{True Value}} \right| \times 100$$
$$= \left| \frac{29.674 - 30.474}{29.674} \right| \times 100$$
$$= 2.6967\%$$

Backward Difference Approximation of the First Derivative

We know

$$f'(x) = \lim_{\Delta x \rightarrow 0} \frac{f(x + \Delta x) - f(x)}{\Delta x}$$

For a finite ' Δx ',

$$f'(x) \approx \frac{f(x + \Delta x) - f(x)}{\Delta x}$$

If ' Δx ' is chosen as a negative number,

$$\begin{aligned} f'(x) &\approx \frac{f(x - \Delta x) - f(x)}{-\Delta x} \\ &= \frac{f(x) - f(x - \Delta x)}{\Delta x} \end{aligned}$$

Backward Difference Approximation of the First Derivative Cont.

This is a backward difference approximation as you are taking a point backward from x . To find the value of $f'(x)$ at $x = x_i$, we may choose another point ' Δx ' behind as $x = x_{i-1}$. This gives

$$\begin{aligned} f'(x_i) &\approx \frac{f(x_i) - f(x_{i-1})}{\Delta x} \\ &= \frac{f(x_i) - f(x_{i-1})}{x_i - x_{i-1}} \end{aligned}$$

where

$$\Delta x = x_i - x_{i-1}$$

Backward Difference Approximation of the First Derivative Cont.

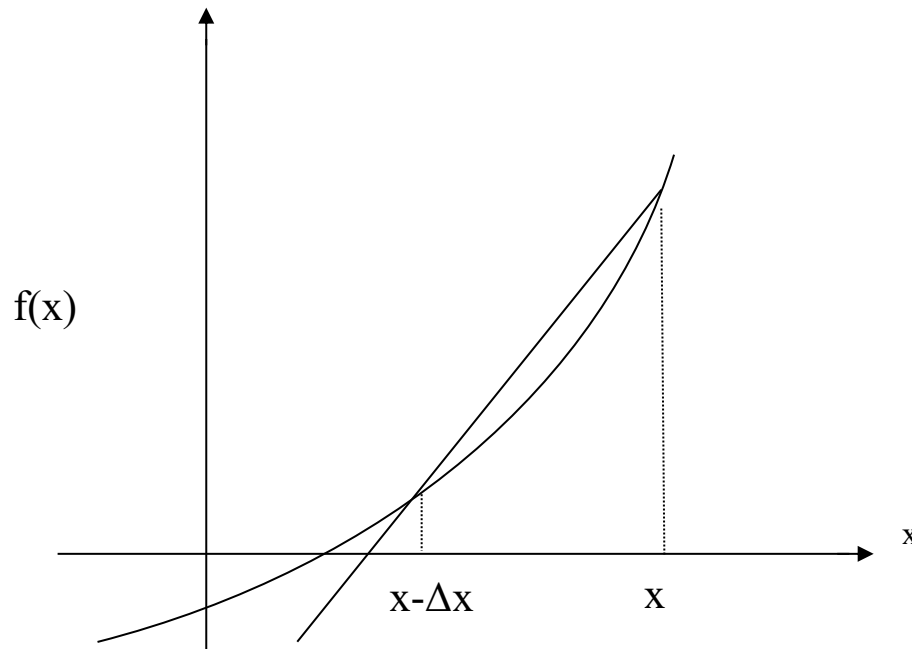


Figure 2 Graphical Representation of backward difference approximation of first derivative

Example 2

The velocity of a rocket is given by

$$v(t) = 2000 \ln \left[\frac{14 \times 10^4}{14 \times 10^4 - 2100t} \right] - 9.8t, 0 \leq t \leq 30$$

where ' v ' is given in m/s and ' t ' is given in seconds.

- Use backward difference approximation of the first derivative of $v(t)$ to calculate the acceleration at $t = 16$ s . Use a step size of $\Delta t = 2$ s .
- Find the absolute relative true error for part (a).

Example 2 Cont.

Solution

$$a(t) \approx \frac{v(t_i) - v(t_{i-1})}{\Delta t}$$

$$t_i = 16$$

$$\Delta t = 2$$

$$\begin{aligned} t_{i-1} &= t_i - \Delta t \\ &= 16 - 2 \\ &= 14 \end{aligned}$$

$$a(16) \approx \frac{v(16) - v(14)}{2}$$

Example 2 Cont.

$$\begin{aligned}v(16) &= 2000 \ln \left[\frac{14 \times 10^4}{14 \times 10^4 - 2100(16)} \right] - 9.8(16) \\&= 392.07 \text{ m/s}\end{aligned}$$

$$\begin{aligned}v(14) &= 2000 \ln \left[\frac{14 \times 10^4}{14 \times 10^4 - 2100(14)} \right] - 9.8(14) \\&= 334.24 \text{ m/s}\end{aligned}$$

$$\begin{aligned}a(16) &\approx \frac{v(16) - v(14)}{2} \\&= \frac{392.07 - 334.24}{2} \\&\approx 28.915 \text{ m/s}^2\end{aligned}$$

Example 2 Cont.

The exact value of the acceleration at $t = 16$ s from Example 1 is

$$a(16) = 29.674 \text{ m/s}^2$$

The absolute relative true error is

$$\begin{aligned} |\epsilon_t| &= \left| \frac{29.674 - 28.915}{29.674} \right| \times 100 \\ &= 2.5584\% \end{aligned}$$

Central Divided Difference

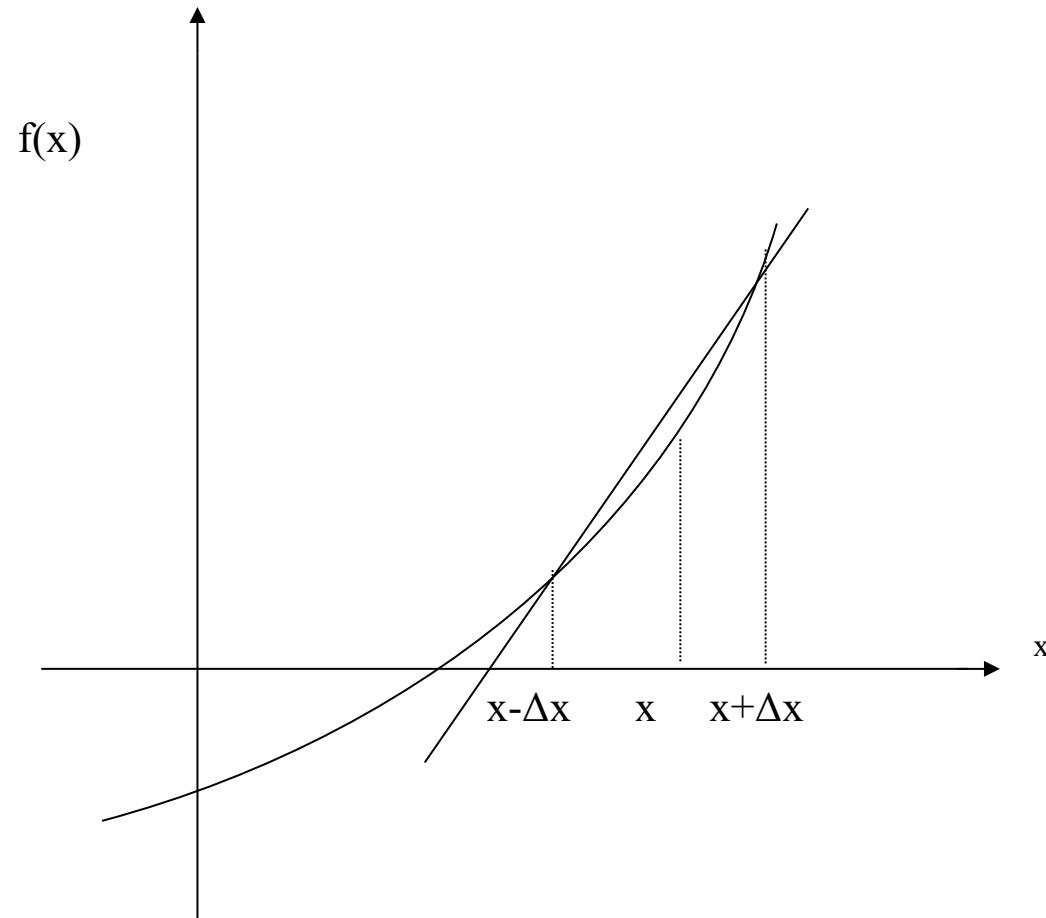


Figure 3 Graphical Representation of central difference approximation of first derivative

Example 3

The velocity of a rocket is given by

$$v(t) = 2000 \ln \left[\frac{14 \times 10^4}{14 \times 10^4 - 2100t} \right] - 9.8t, 0 \leq t \leq 30$$

where ' v ' is given in m/s and ' t ' is given in seconds.

- (a) Use central divided difference approximation of the first derivative of $v(t)$ to calculate the acceleration at $t = 16s$. Use a step size of $\Delta t = 2s$.
- (b) Find the absolute relative true error for part (a).

Example 3 cont.

Solution

$$a(t_i) \approx \frac{v(t_{i+1}) - v(t_{i-1}))}{2\Delta t}$$

$$t_i = 16$$

$$\Delta t = 2$$

$$\begin{aligned} t_{i+1} &= t_i + \Delta t \\ &= 16 + 2 \\ &= 18 \end{aligned}$$

$$\begin{aligned} t_{i-1} &= t_i - \Delta t \\ &= 16 - 2 \\ &= 14 \end{aligned}$$

$$\begin{aligned} a(16) &\approx \frac{v(18) - v(14)}{2(2)} \\ &\approx \frac{v(18) - v(14)}{4} \end{aligned}$$

Example 3 cont.

$$\begin{aligned}\nu(18) &= 2000 \ln \left[\frac{14 \times 10^4}{14 \times 10^4 - 2100(18)} \right] - 9.8(18) \\ &= 453.02 \text{ m/s}\end{aligned}$$

$$\begin{aligned}\nu(14) &= 2000 \ln \left[\frac{14 \times 10^4}{14 \times 10^4 - 2100(14)} \right] - 9.8(14) \\ &= 334.24 \text{ m/s}\end{aligned}$$

$$\begin{aligned}a(16) &\approx \frac{\nu(18) - \nu(14)}{4} \\ &\approx \frac{453.02 - 334.24}{4} \\ &\approx 29.694 \text{ m/s}^2\end{aligned}$$

Example 3 cont.

The exact value of the acceleration at $t = 16$ s from Example 1 is

$$a(16) = 29.674 \text{ m/s}^2$$

The absolute relative true error is

$$\begin{aligned} |\epsilon_t| &= \left| \frac{29.674 - 29.694}{29.674} \right| \times 100 \\ &= 0.069157 \% \end{aligned}$$

Comparision of FDD, BDD, CDD

The results from the three difference approximations are given in Table 1.

Table 1 Summary of $a(16)$ using different divided difference approximations

Type of Difference Approximation	$a(16)$ (m/s^2)	$ \epsilon_t \%$
Forward	30.475	2.6967
Backward	28.915	2.5584
Central	29.695	0.069157

Finding the value of the derivative within a prespecified tolerance

In real life, one would **not know the exact value** of the derivative – so how would one know how accurately they have found the value of the derivative.

A simple way would be to **start with a step size** and **keep on halving** the step size and keep on halving the step size until the absolute relative approximate error is within a pre-specified tolerance.

Take the example of finding $v'(t)$ for

$$v(t) = 2000 \ln \left[\frac{14 \times 10^4}{14 \times 10^4 - 2100t} \right] - 9.8t$$

at $t = 16$ using the backward divided difference scheme.

Finding the value of the derivative within a prespecified tolerance Cont.

Given in Table 2 are the values obtained using the backward difference approximation method and the corresponding absolute relative approximate errors.

Table 2 First derivative approximations and relative errors for different Δt values of backward difference scheme

Δt	$v'(t)$	$ \epsilon_a \%$
2	28.915	
1	29.289	1.2792
0.5	29.480	0.64787
0.25	29.577	0.32604
0.125	29.625	0.16355

Finding the value of the derivative within a prespecified tolerance Cont.

From the above table, one can see that the absolute relative approximate error decreases as the step size is reduced. At $\Delta t = 0.125$ the absolute relative approximate error is 0.16355%, meaning that at least 2 significant digits are correct in the answer.

Differentiation-Discrete Functions

Example 1

The upward velocity of a rocket is given as a function of time in Table 1.

Table 1 Velocity as a function of time

t	v(t)
s	m/s
0	0
10	227.04
15	362.78
20	517.35
22.5	602.97
30	901.67

Using forward divided difference, find the acceleration of the rocket at $t = 16$ s

Example 1 Cont.

Solution

To find the acceleration at $t = 16\text{s}$, we need to choose the two values closest to $t = 16\text{s}$, that also bracket $t = 16\text{s}$ to evaluate it. The two points are $t = 15\text{s}$ and $t = 20\text{s}$

$$a(t_i) \approx \frac{v(t_{i+1}) - v(t_i)}{\Delta t}$$

$$t_i = 15$$

$$t_{i+1} = 20$$

$$\begin{aligned}\Delta t &= t_{i+1} - t_i \\ &= 20 - 15 \\ &= 5\end{aligned}$$

Example 1 Cont.

$$\begin{aligned}a(16) &\approx \frac{v(20) - v(15)}{5} \\ &\approx \frac{517.35 - 362.78}{5} \\ &\approx 30.914 \text{ m/s}^2\end{aligned}$$

Direct Fit Polynomials

In this method, given ' $n + 1$ ' data points $(x_0, y_0), (x_1, y_1), (x_2, y_2), \dots, (x_n, y_n)$ one can fit a n^{th} order polynomial given by

$$P_n(x) = a_0 + a_1x + \dots + a_{n-1}x^{n-1} + a_nx^n$$

To find the first derivative,

$$P'_n(x) = \frac{dP_n(x)}{dx} = a_1 + 2a_2x + \dots + (n-1)a_{n-1}x^{n-2} + na_nx^{n-1}$$

Similarly other derivatives can be found.

Example 2-Direct Fit Polynomials

The upward velocity of a rocket is given as a function of time in Table 2.

Table 2 Velocity as a function of time

t	v(t)
s	m/s
0	0
10	227.04
15	362.78
20	517.35
22.5	602.97
30	901.67

Using the third order polynomial interpolant for velocity, find the acceleration of the rocket at $t = 16$ s.

Example 2-Direct Fit Polynomials cont.

Solution

For the third order polynomial (also called cubic interpolation), we choose the velocity given by

$$v(t) = a_0 + a_1t + a_2t^2 + a_3t^3$$

Since we want to find the velocity at $t = 16$ s and we are using third order polynomial, we need to choose the four points closest to $t = 16$ s and that also bracket $t = 16$ s to evaluate it.

The four points are $t_o = 10$, $t_1 = 15$, $t_2 = 20$, and $t_3 = 22.5$.

$$t_o = 10, \quad v(t_o) = 227.04$$

$$t_1 = 15, \quad v(t_1) = 362.78$$

$$t_2 = 20, \quad v(t_2) = 517.35$$

$$t_3 = 22.5, \quad v(t_3) = 602.97$$

Example 2-Direct Fit Polynomials cont.

such that

$$v(10) = 227.04 = a_0 + a_1(10) + a_2(10)^2 + a_3(10)^3$$

$$v(15) = 362.78 = a_0 + a_1(15) + a_2(15)^2 + a_3(15)^3$$

$$v(20) = 517.35 = a_0 + a_1(20) + a_2(20)^2 + a_3(20)^3$$

$$v(22.5) = 602.97 = a_0 + a_1(22.5) + a_2(22.5)^2 + a_3(22.5)^3$$

Example 2-Direct Fit Polynomials cont.

Solving the above four equations gives

$$a_0 = -4.3810$$

$$a_1 = 21.289$$

$$a_2 = 0.13065$$

$$a_3 = 0.0054606$$

Hence

$$\begin{aligned} v(t) &= a_0 + a_1 t + a_2 t^2 + a_3 t^3 \\ &= -4.3810 + 21.289t + 0.13065t^2 + 0.0054606t^3, \quad 10 \leq t \leq 22.5 \end{aligned}$$

Example 2-Direct Fit Polynomials cont.

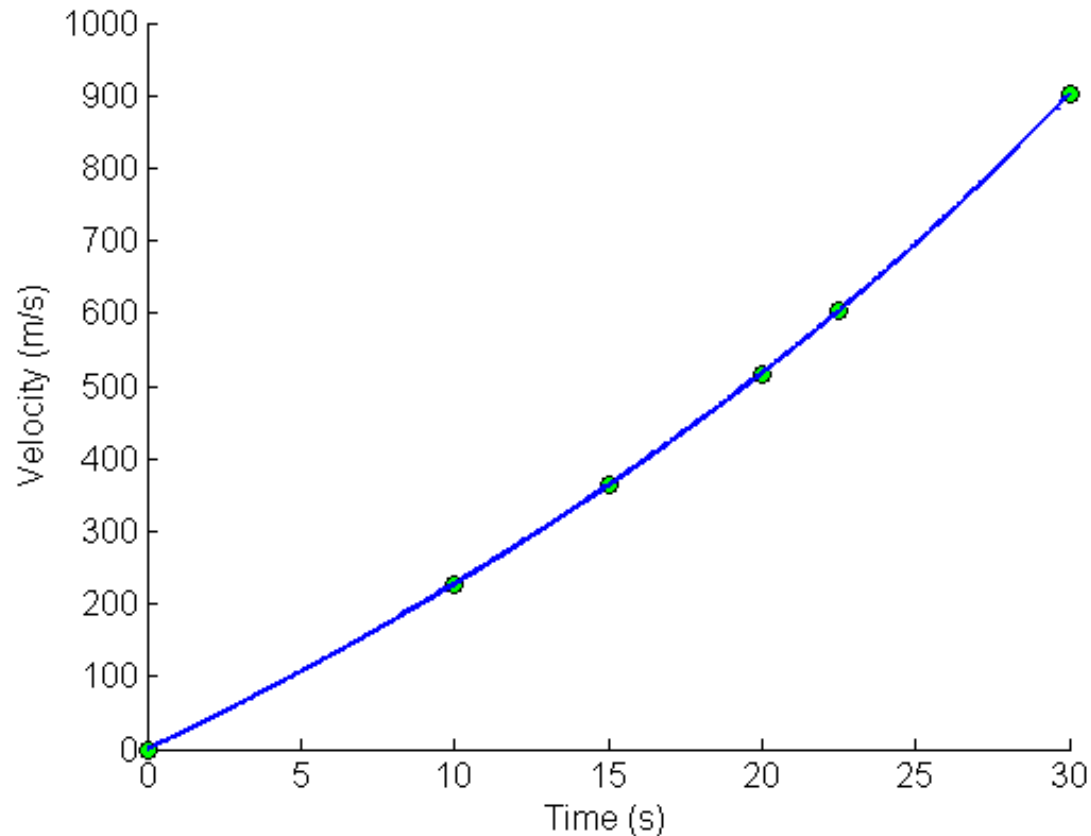


Figure 1 Graph of upward velocity of the rocket vs. time.

Example 2-Direct Fit Polynomials cont.

The acceleration at $t=16$ is given by

$$a(16) = \left. \frac{d}{dt} v(t) \right|_{t=16}$$

Given that

$$v(t) = -4.3810 + 21.289t + 0.13065t^2 + 0.0054606t^3, 10 \leq t \leq 22.5$$

$$\begin{aligned} a(t) &= \frac{d}{dt} v(t) \\ &= \frac{d}{dt} (-4.3810 + 21.289t + 0.13065t^2 + 0.0054606t^3) \\ &= 21.289 + 0.26130t + 0.016382t^2, \quad 10 \leq t \leq 22.5 \end{aligned}$$

$$\begin{aligned} a(16) &= 21.289 + 0.26130(16) + 0.016382(16)^2 \\ &= 29.664 \text{ m/s}^2 \end{aligned}$$

Lagrange Polynomial

$$f_n(x) = \sum_{i=0}^n L_i(x) f(x_i)$$

where ' n ' in $f_n(x)$ stands for the n^{th} order polynomial that approximates the function $y = f(x)$ given at $(n + 1)$ data points as $(x_0, y_0), (x_1, y_1), \dots, (x_{n-1}, y_{n-1}), (x_n, y_n)$, and

$$L_i(x) = \prod_{\substack{j=0 \\ j \neq i}}^n \frac{x - x_j}{x_i - x_j}$$

$L_i(x)$ a weighting function that includes a product of $(n - 1)$ terms with terms of $j = i$ omitted.

Lagrange Polynomial Cont.

Then to find the first derivative, one can differentiate $f_n(x)$ once, and so on for other derivatives.

$$f_n(x) = \sum_{i=0}^n L_i(x) f(x_i)$$

For example, the second order Lagrange polynomial passing through

$(x_0, y_0), (x_1, y_1), (x_2, y_2)$ is

$$f_2(x) = \boxed{}$$

Differentiating equation (2) gives

$$L_i(x) = \prod_{\substack{j=0 \\ j \neq i}}^n \frac{x - x_j}{x_i - x_j}$$

Lagrange Polynomial Cont.

$$f_2'(x) = \frac{2x - (x_1 + x_2)}{(x_0 - x_1)(x_0 - x_2)} f(x_0) + \frac{2x - (x_0 + x_2)}{(x_1 - x_0)(x_1 - x_2)} f(x_1) + \frac{2x - (x_0 + x_1)}{(x_2 - x_0)(x_2 - x_1)} f(x_2)$$

Differentiating again would give the second derivative as

$$f_2''(x) = \frac{2}{(x_0 - x_1)(x_0 - x_2)} f(x_0) + \frac{2}{(x_1 - x_0)(x_1 - x_2)} f(x_1) + \frac{2}{(x_2 - x_0)(x_2 - x_1)} f(x_2)$$

Example 3

The upward velocity of a rocket is given as a function of time in Table 3.

Table 3 Velocity as a function of time

t	v(t)
s	m/s
0	0
10	227.04
15	362.78
20	517.35
22.5	602.97
30	901.67

Determine the value of the acceleration at $t = 16$ s using the second order Lagrangian polynomial interpolation for velocity.

Example 3 Cont.

$$t_0=10, \quad t_1=15, \quad t_2=20, \quad t=16$$

Solution

$$v(t) = \left(\frac{t-t_1}{t_0-t_1} \right) \left(\frac{t-t_2}{t_0-t_2} \right) v(t_0) + \left(\frac{t-t_0}{t_1-t_0} \right) \left(\frac{t-t_2}{t_1-t_2} \right) v(t_1) + \left(\frac{t-t_0}{t_2-t_0} \right) \left(\frac{t-t_1}{t_2-t_1} \right) v(t_2)$$

$$a(t) = \frac{2t-(t_1+t_2)}{(t_0-t_1)(t_0-t_2)} v(t_0) + \frac{2t-(t_0+t_2)}{(t_1-t_0)(t_1-t_2)} v(t_1) + \frac{2t-(t_0+t_1)}{(t_2-t_0)(t_2-t_1)} v(t_2)$$

$$\begin{aligned} a(16) &= \frac{2(16)-(15+20)}{(10-15)(10-20)} (227.04) + \frac{2(16)-(10+20)}{(15-10)(15-20)} (362.78) + \frac{2(16)-(10+15)}{(20-10)(20-15)} (517.35) \\ &= -0.06(227.04) - 0.08(362.78) + 0.14(517.35) \\ &= 29.784 \text{ m/s}^2 \end{aligned}$$

Trapezoidal Rule of Integration

What is Integration

Integration:

The process of measuring the area under a function plotted on a graph.

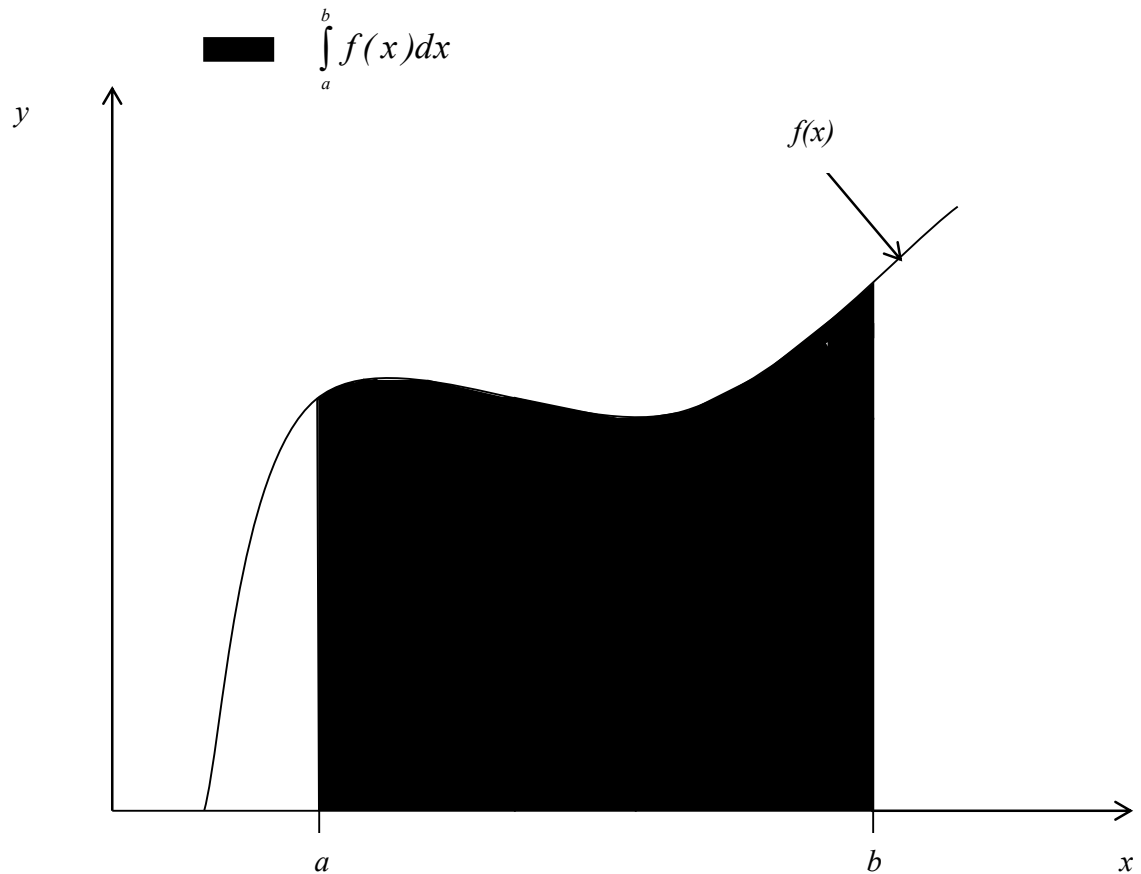
$$I = \int_a^b f(x) dx$$

Where:

$f(x)$ is the integrand

a = lower limit of integration

b = upper limit of integration



Basis of Trapezoidal Rule

Trapezoidal Rule is based on the Newton-Cotes Formula that states if one can **approximate the integrand as an n^{th} order polynomial...**

$$I = \int_a^b f(x) dx \quad \text{where} \quad f(x) \approx f_n(x)$$

$$\text{and} \quad f_n(x) = a_0 + a_1x + \dots + a_{n-1}x^{n-1} + a_nx^n$$

Basis of Trapezoidal Rule

Then the integral of that function is approximated by the integral of that n^{th} order polynomial.

$$\int_a^b f(x) \approx \int_a^b f_n(x)$$

Trapezoidal Rule assumes **$n=1$** , that is, the area under the **linear polynomial**,

$$\int_a^b f(x)dx = (b-a) \left[\frac{f(a) + f(b)}{2} \right]$$

Method Derived From Geometry

The area under the curve is a **trapezoid**. The integral

$$\begin{aligned} \int_a^b f(x)dx &\approx \text{Area of trapezoid} \\ &= \frac{1}{2}(\text{Sum of parallel sides})(\text{height}) \\ &= \frac{1}{2}(f(b) + f(a))(b - a) \\ &= (b - a) \left[\frac{f(a) + f(b)}{2} \right] \end{aligned}$$

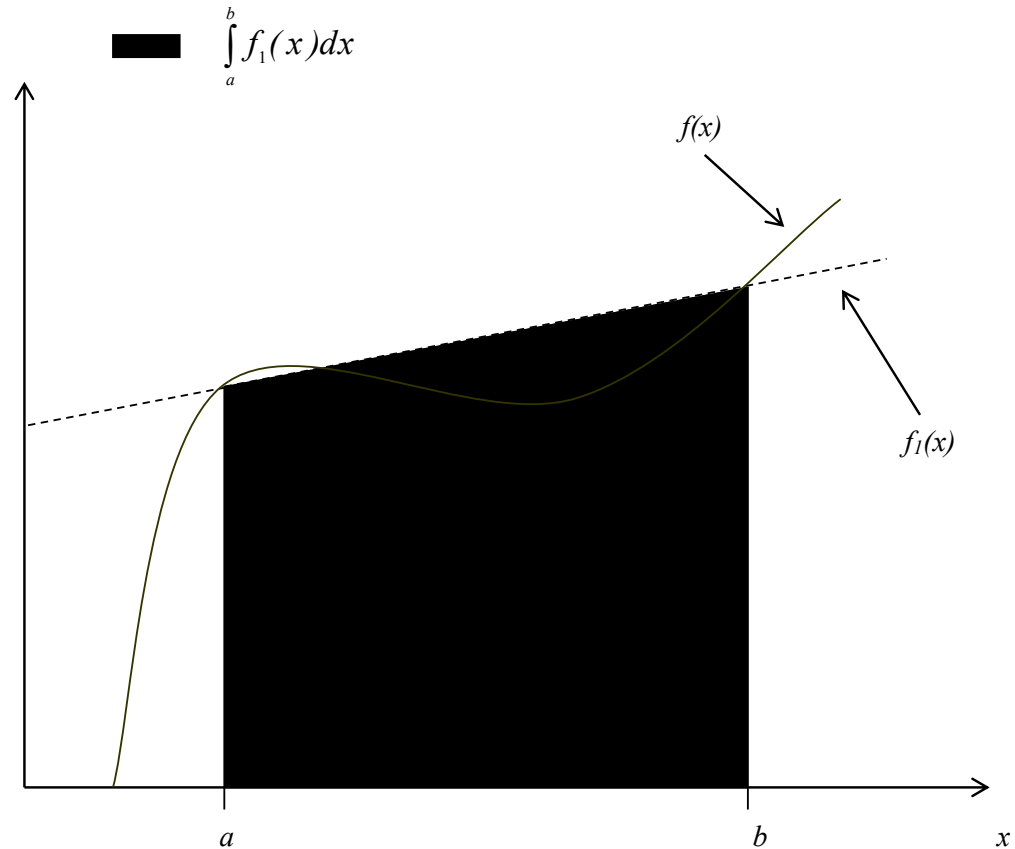


Figure 2: Geometric Representation

Example 1

The vertical distance covered by a rocket from $t=8$ to $t=30$ seconds is given by:

$$x = \int_8^{30} \left(2000 \ln \left[\frac{140000}{140000 - 2100t} \right] - 9.8t \right) dt$$

- a) Use single segment Trapezoidal rule to find the distance covered.
- b) Find the true error, E_t for part (a).
- c) Find the absolute relative true error, $|\epsilon_a|$ for part (a).

Solution

$$\text{a)} \quad I \approx (b - a) \left[\frac{f(a) + f(b)}{2} \right]$$

$$a = 8 \quad b = 30$$

$$f(t) = 2000 \ln \left[\frac{140000}{140000 - 2100t} \right] - 9.8t$$

$$f(8) = 2000 \ln \left[\frac{140000}{140000 - 2100(8)} \right] - 9.8(8) = 177.27 \text{ m/s}$$

$$f(30) = 2000 \ln \left[\frac{140000}{140000 - 2100(30)} \right] - 9.8(30) = 901.67 \text{ m/s}$$

Solution (cont)

$$I \approx (b - a) \left[\frac{f(a) + f(b)}{2} \right]$$

a)

$$I = (30 - 8) \left[\frac{177.27 + 901.67}{2} \right]$$

$$= 11868 \text{ m}$$

b) The exact value of the above integral is

$$x = \int_8^{30} \left(2000 \ln \left[\frac{140000}{140000 - 2100t} \right] - 9.8t \right) dt = 11061 \text{ m}$$

Solution (cont)

b)

$$\begin{aligned} E_t &= \text{True Value} - \text{Approximate Value} \\ &= 11061 - 11868 \\ &= -807 \text{ m} \end{aligned}$$

c) The absolute relative true error, $|\epsilon_t|$, would be

$$|\epsilon_t| = \left| \frac{11061 - 11868}{11061} \right| \times 100 = 7.2959\%$$

Multiple Segment Trapezoidal Rule

In Example 1, the true error using **single segment** trapezoidal rule was **large**.

We can **divide** the **interval** [8,30] into [8,19] and [19,30] intervals and apply Trapezoidal rule over each segment.

$$f(t) = 2000 \ln \left(\frac{140000}{140000 - 2100t} \right) - 9.8t$$

$$\int_8^{30} f(t) dt = \int_8^{19} f(t) dt + \int_{19}^{30} f(t) dt$$

$$= (19 - 8) \left[\frac{f(8) + f(19)}{2} \right] + (30 - 19) \left[\frac{f(19) + f(30)}{2} \right]$$

Multiple Segment Trapezoidal Rule

With $f(8) = 177.27 \text{ m/s}$
 $f(30) = 901.67 \text{ m/s}$
 $f(19) = 484.75 \text{ m/s}$

Hence:

$$= (19 - 8) \left[\frac{f(8) + f(19)}{2} \right] + (30 - 19) \left[\frac{f(19) + f(30)}{2} \right]$$

$$\int_8^{30} f(t) dt = (19 - 8) \left[\frac{177.27 + 484.75}{2} \right] + (30 - 19) \left[\frac{484.75 + 901.67}{2} \right]$$
$$= 11266 \text{ m}$$

Multiple Segment Trapezoidal Rule

The true error is:

$$\begin{aligned} E_t &= 11061 - 11266 \\ &= -205 \text{ m} \end{aligned}$$

The true error now is reduced from -807 m to -205 m.

Extending this procedure to divide the interval into equal segments to apply the Trapezoidal rule;

the sum of the results obtained for each segment is the approximate value of the integral.

Multiple Segment Trapezoidal Rule

Divide into equal segments as shown in Figure 4. Then the width of each segment is:

$$h = \frac{b - a}{n}$$

The integral I is:

$$I = \int_a^b f(x) dx$$

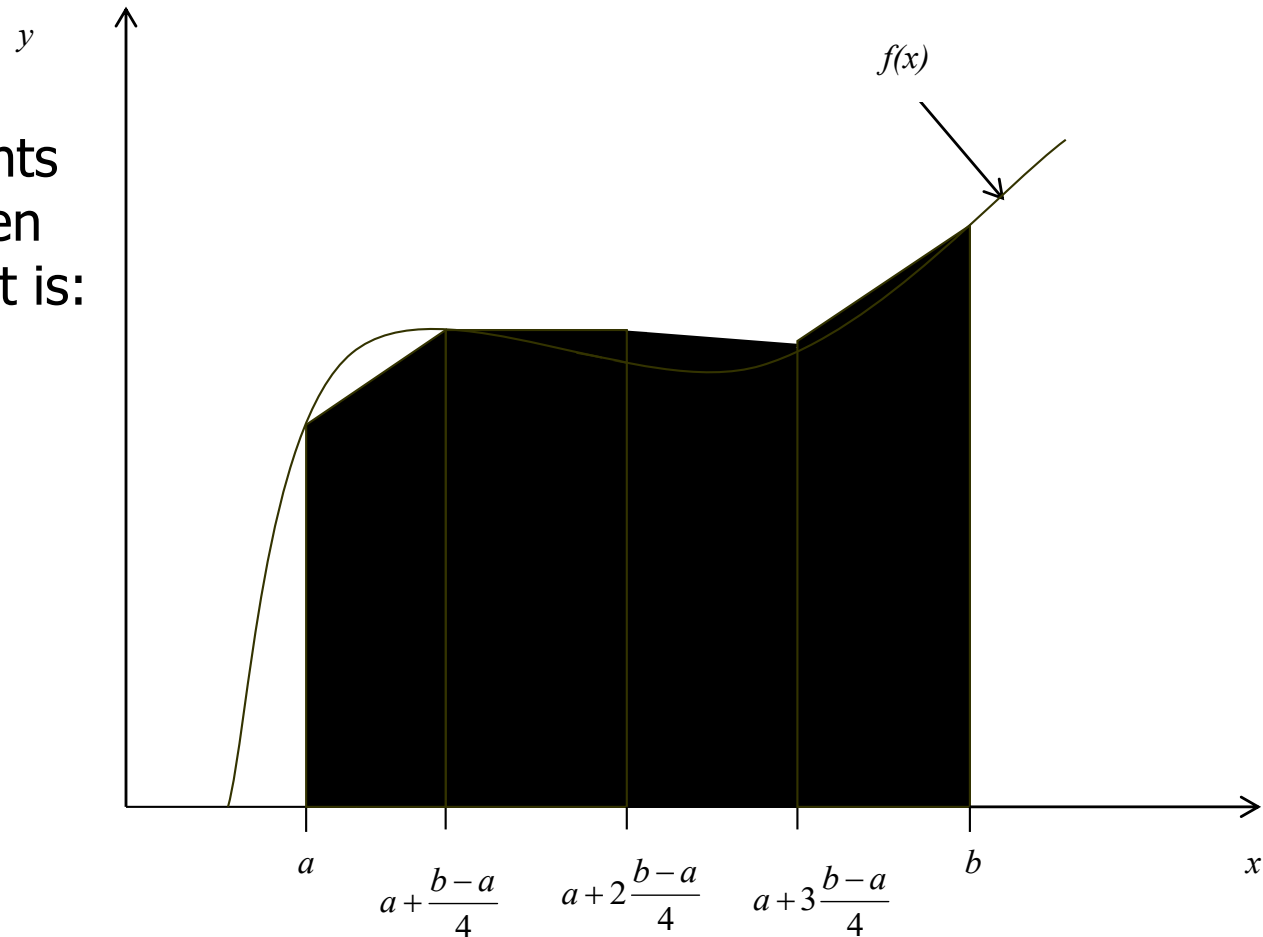


Figure 4: Multiple (n=4) Segment Trapezoidal Rule

Multiple Segment Trapezoidal Rule

The integral I can be broken into h integrals as:

$$\int_a^b f(x)dx = \int_a^{a+h} f(x)dx + \int_{a+h}^{a+2h} f(x)dx + \dots + \int_{a+(n-2)h}^{a+(n-1)h} f(x)dx + \int_{a+(n-1)h}^b f(x)dx$$

Applying Trapezoidal rule on each segment gives:

$$I \approx (b-a) \left[\frac{f(a) + f(b)}{2} \right] \qquad h = \frac{b-a}{n}$$

$$\int_a^b f(x)dx = \frac{b-a}{2n} \left[f(a) + 2 \left\{ \sum_{i=1}^{n-1} f(a+ih) \right\} + f(b) \right]$$

Example 2

The vertical distance covered by a rocket from $t = 8$ to $t = 30$ seconds is given by:

$$x = \int_8^{30} \left(2000 \ln \left[\frac{140000}{140000 - 2100t} \right] - 9.8t \right) dt$$

- a) Use two-segment Trapezoidal rule to find the distance covered.
- b) Find the true error, E_t for part (a).
- c) Find the absolute relative true error, $|\epsilon_a|$ for part (a).

Solution

a) The solution using 2-segment Trapezoidal rule is

$$I = \frac{b-a}{2n} \left[f(a) + 2 \left\{ \sum_{i=1}^{n-1} f(a+ih) \right\} + f(b) \right]$$

$$n = 2 \qquad a = 8 \qquad b = 30$$

$$h = \frac{b-a}{n} = \frac{30-8}{2} = 11$$

Solution (cont)

$$n = 2 \qquad a = 8 \qquad b = 30$$

$$h = \frac{b-a}{n} = \frac{30-8}{2} = 11$$

Then:

$$I = \frac{b-a}{2n} \left[f(a) + 2 \left\{ \sum_{i=1}^{n-1} f(a+ih) \right\} + f(b) \right]$$

$$I = \frac{30-8}{2(2)} \left[f(8) + 2 \left\{ \sum_{i=1}^{2-1} f(a+ih) \right\} + f(30) \right]$$

$$= \frac{22}{4} [f(8) + 2f(19) + f(30)]$$

$$= \frac{22}{4} [177.27 + 2(484.75) + 901.67]$$

$$= 11266 \text{ m}$$

Solution (cont)

b) The exact value of the above integral is

$$x = \int_8^{30} \left(2000 \ln \left[\frac{140000}{140000 - 2100t} \right] - 9.8t \right) dt = 11061 \text{ m}$$

so the true error is

$$E_t = \text{True Value} - \text{Approximate Value}$$

$$= 11061 - 11266$$

Solution (cont)

The absolute relative true error, $|\epsilon_t|$, would be

$$|\epsilon_t| = \left| \frac{\text{True Error}}{\text{True Value}} \right| \times 100$$

$$= \left| \frac{11061 - 11266}{11061} \right| \times 100$$

$$= 1.8534\%$$

Solution (cont)

Table 1 gives the values obtained using multiple segment Trapezoidal rule for

$$x = \int_8^{30} \left(2000 \ln \left[\frac{140000}{140000 - 2100t} \right] - 9.8t \right) dt$$

Exact Value=11061 m

n	Value	E_t	 ε_t %	 ε_a %
1	11868	-807	7.296	---
2	11266	-205	1.853	5.343
3	11153	-91.4	0.8265	1.019
4	11113	-51.5	0.4655	0.3594
5	11094	-33.0	0.2981	0.1669
6	11084	-22.9	0.2070	0.09082
7	11078	-16.8	0.1521	0.05482
8	11074	-12.9	0.1165	0.03560

Table 1: Multiple Segment Trapezoidal Rule Values

Activity

Use Multiple Segment Trapezoidal Rule to find the area under the curve

$$f(x) = \frac{300x}{1 + e^x} \quad \text{from } x = 0 \quad \text{to} \quad x = 10$$

Using two segments, we get $h = \frac{10 - 0}{2} = 5$ and

$$f(0) = \frac{300(0)}{1 + e^0} = 0 \quad f(5) = \frac{300(5)}{1 + e^5} = 10.039 \quad f(10) = \frac{300(10)}{1 + e^{10}} = 0.136$$